

Leveraging Drinking Water Pumps as Flexible Loads Using Input Convex Neural Networks

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Abstract—Drinking water distribution networks can be operated as flexible loads within the electric power grid due to their substantial pumping demands and water storage capabilities. Optimizing the flexible operation of a water distribution network poses significant challenges due to the complex physical laws within the network, where the hydraulic head difference equations for pipes and pumps are nonconvex. Standard nonconvex optimization solvers often fail to provide globally optimal solutions and the time required for computation can be prohibitively large. To resolve these issues, we present an optimization approach that accurately approximates the nonconvex constraints using input convex neural networks (ICNNs). This method converts the mixed-integer nonconvex optimization problem into a mixed-integer linear program, improving computational efficiency and scalability while maintaining the optimization problem's intuitive structure. In two case studies, we compare the ICNN-aided approach with the original nonconvex problem and found that the ICNN-aided approach outperforms the nonconvex solver in terms of computational time and optimality.

Index Terms—Flexible loads, neural networks, optimization, water distribution networks, water pumping

I. INTRODUCTION

Flexibility is becoming increasingly important within the electric power grid due to the increase of renewable energy sources. Since renewable energy sources—such as solar and wind—are intermittent and uncertain, more flexibility is needed to balance mismatches in supply and demand. Flexible loads can help provide flexibility by shifting or reducing their power consumption and are widely recognized as assets capable of responding within price-based or incentive-based demand response programs, which reflect changes in real-time grid needs [1].

Drinking water distribution networks can be operated as flexible loads within the power grid because of their substantial pumping demands and water storage capabilities [2]. Prioritizing pumping during off-peak periods helps reduce costs, balance the grid, and utilize surplus renewable energy, thus supporting grid reliability. Unfortunately, the optimal water pumping problem is subject to the water flow constraints which are inherently nonconvex. This yields a mixed-integer nonconvex program, where nonconvex optimization solvers often fail to provide globally optimal (or even feasible) solutions, and the time required for computation can be prohibitively large.

In the literature, approaches to solve the optimal water pumping problem have been investigated in [3]–[7]. An iterative mixed-integer linear approximation is developed in [3] and a second-order cone relaxation is developed in [4]. Ref. [5] proposes a chance-constrained optimization framework to account for water and power demand uncertainty and is solved

with a combination of relaxations and approximations. However, all of these formulations assume that the pumps are fixed speed. Water system operators are increasingly adopting variable speed pumps due to their improved efficiency and control. In [6], the variable speed pump head difference equation is approximated but the pump power consumption is not considered. Ref. [7] develops a piece-wise linearization of the pipe and pump equations, where a binary variable is introduced for each piece-wise segment of the pumps and pipes for all time periods. However, issues remain whether these solution approaches can generalize across different water distribution networks and for larger networks.

In this work, we investigate the use of input convex neural networks (ICNNs) to develop a computationally tractable water pumping problem. ICNNs were proposed in [8], where the neural network includes additional requirements on the network parameters to ensure that the output is a convex function of the inputs. ICNNs will allow us to approximate the nonconvex head difference equations with a set of linear constraints. The resulting formulation is a mixed-integer linear program, which can be more readily solved with commercial solvers. In [9], ICNNs were applied to the gas network to approximate the Weymouth equations. In this paper, we extend ICNNs to apply to the optimal water pumping problem and investigate impact of the approximation on the solution performance and accuracy. To do this, we investigate two different case studies over multiple periods.

The contributions of this paper are i) formulating the multiperiod optimal water pumping problem subject to the water flow constraints while minimizing the real-time electricity costs, ii) developing ICNN-based approximations of the pipe and pump head difference equations, iii) comparing the computational and solution performance of the ICNN-aided approach against the original nonconvex water pumping problem within two case studies, and iv) discussing the impact of this approximation on controlling the water distribution network as a flexible load. The paper is structured as follows: Section II presents an overview of the nonconvex optimal water pumping problem and how the ICNN approach is integrated into this framework. Section III presents two case studies to investigate the performance of the proposed approach. Last, final observations and insights are presented in Section IV.

II. PROBLEM DESCRIPTION

The optimal water pumping problem minimizes the electricity costs associated with operating water distribution network pumps over a scheduling horizon. The power consumption of

the pumps is determined by the pump status and speed setting. The solution must satisfy the water flow constraints in order to meet consumer water demands as well as maintain sufficient water levels in tanks and nodal pressure requirements. The optimal water pumping problem is of the following form:

$$\begin{aligned} & \underset{\mathbf{X}}{\text{minimize}} && F(\mathbf{X}) \\ & \text{subject to} && g_1(\mathbf{X}) = 0, \\ & && g_2(\mathbf{X}) \leq 0, \end{aligned} \quad (\text{P1})$$

where the cost function $F(\mathbf{X})$ represents the electricity costs given the decision variable vector $\mathbf{X}$. The functions $g_1(\cdot)$ and $g_2(\cdot)$ represent the water flow constraints and limits, which are described in Section II-A. The complete nonconvex water pumping problem is presented in Section II-B. Next, the ICNN approximation approach is developed in Section II-C and the mixed-integer linear programming approximation is presented in Section II-D.

A. Water Flow Constraints

We model the water distribution network as a directed graph consisting of sets of edges $\mathcal{E}$ and nodes $\mathcal{N}$. We formulate the water pumping problem as a multiperiod problem, considering all time steps t within the scheduling horizon $\mathcal{T}$, with periods of duration ΔT . We denote the hydraulic head at node j and time t as H_j^t and the volumetric flow rate in edge ij and time t as x_{ij}^t .

1) *Edges*: The edges are either pumps $\mathcal{P}$ or pipes $\mathcal{W}$, i.e., $\mathcal{E} = \mathcal{P} \cup \mathcal{W}$. For pumps, we can control the normalized pump speed w_{ij}^t and the binary pump status s_{ij}^t (i.e., the pump is on when $s_{ij}^t = 1$). We model the head difference along pipes and pumps as well as the performance of the pumps for all $t \in \mathcal{T}$

$$\underline{w}_{ij} \cdot s_{ij}^t \leq w_{ij}^t \leq \bar{w}_{ij} \cdot s_{ij}^t \quad \forall ij \in \mathcal{P}, \quad (1)$$

$$0 \leq x_{ij}^t \leq \bar{x}_{ij} \cdot s_{ij}^t \quad \forall ij \in \mathcal{P}, \quad (2)$$

$$s_{ij}^t = 1 \Rightarrow H_j^t - H_i^t = b_{ij}^0 (w_{ij}^t)^2 + b_{ij}^1 (x_{ij}^t)^2 \quad \forall ij \in \mathcal{P}, \quad (3)$$

$$p_{ij}^t = c_{ij}^0 \cdot (w_{ij}^t)^2 + c_{ij}^1 \cdot x_{ij}^t \quad \forall ij \in \mathcal{P}, \quad (4)$$

$$s_{ij}^t \in \{0, 1\} \quad \forall ij \in \mathcal{P}, \quad (5)$$

$$H_i^t - H_j^t = k_{ij} \cdot x_{ij}^t \cdot |x_{ij}^t| \quad \forall ij \in \mathcal{W}, \quad (6)$$

where $\underline{w}_{ij}$ and $\bar{w}_{ij}$ are the minimum and maximum speed of pump ij and $\bar{x}_{ij}$ is the maximum flow rate of pump ij . The pump speed is bounded in (1) to stay within $\underline{w}_{ij}$ and $\bar{w}_{ij}$ when the pump is on. When the pump is off, the pump speed is forced to zero. In (2), the water flow rate through pump ij has an upper bound of $\bar{x}_{ij}$ when the pump is on. When the pump is off, the flow rate is forced to zero. The hydraulic head gain of pump ij is represented by (3), where b_{ij}^0 and b_{ij}^1 are the nominal pump curve parameters. An indicator constraint is used since the head gain equation only holds when the pump is on. The pump power p_{ij}^t is dependent on the flow rate and speed, and can be approximated by (4), where c_{ij}^0 and c_{ij}^1 are parameters of the pump power consumption curve at the nominal pump speed. The hydraulic head loss across the pipes is modeled with the Darcy-Weisbach head loss formula in (6), where k_{ij} is the resistance coefficient of pipe ij .

2) *Nodes*: Each node is classified as either a junction $\mathcal{J}$, reservoir $\mathcal{R}$, or storage tank $\mathcal{S}$, i.e., $\mathcal{N} = \mathcal{J} \cup \mathcal{R} \cup \mathcal{S}$. Junctions connect consumers or multiple pipes, reservoirs are large water sources, and storage tanks store and supply water. The junction, reservoir, and storage tank constraints are

$$\underline{H}_j \leq H_j^t \leq \bar{H}_j \quad \forall j \in \mathcal{J} \cup \mathcal{S}, t \in \mathcal{T}, \quad (7)$$

$$\sum_{i:ij \in \mathcal{E}} x_{ij}^t + d_j^t = 0 \quad \forall j \in \mathcal{J}, t \in \mathcal{T}, \quad (8)$$

$$H_j^t = \hat{h}_j \quad \forall j \in \mathcal{R}, t \in \mathcal{T}, \quad (9)$$

$$H_j^t = H_j^{t-1} + \frac{\Delta T}{\gamma_j} \sum_{i:ij \in \mathcal{E}} x_{ij}^{t-1} \quad \forall j \in \mathcal{S}, t \in \mathcal{T}, \quad (10)$$

$$\sum_{j \in \mathcal{S}} H_j^{Nt} \geq \sum_{j \in \mathcal{S}} H_j^{t=0}. \quad (11)$$

The hydraulic heads at all junctions and tanks are bounded in (7), where $\underline{H}_j$ and $\bar{H}_j$ are the minimum and maximum head limits at node j . Conservation of water throughout the network must be satisfied in (8), where d_j^t is the water demand at junction j and time t . In (9), we model the reservoirs as infinite sources with a fixed hydraulic head $\hat{h}_j$. In (10), the tank level is dependent on the previous period's head and water entering or exiting the tank, where γ_j is the cross-sectional area of tank j . Additionally, we include constraint (11) to ensure that the water stored in the tank at the final time period Nt within the scheduling horizon is maintained at or above the initial volumes.

B. Nonconvex Water Pumping Problem

The optimal water pumping problem solves for status, speed, and power consumption of the pumps subject to the water flow constraints. The optimization problem is then

$$\begin{aligned} & \underset{\mathbf{X}}{\text{minimize}} && \sum_{t \in \mathcal{T}} \sum_{ij \in \mathcal{P}} \pi_{ij}^t \cdot p_{ij}^t \cdot \Delta T \\ & \text{subject to} && (1) - (11). \end{aligned} \quad (\text{NC})$$

where π_{ij}^t is the time-varying electricity price of pump ij at time t . The objective function minimizes the total electricity cost associated with operating the pumps in the water distribution networks over the scheduling horizon. The decision variables are hydraulic head H_j^t , water flow rate x_{ij}^t , pump power p_{ij}^t , pump speed w_{ij}^t , and binary pump status s_{ij}^t for all edges, nodes, and time periods. This is a mixed-integer nonconvex problem due the binary pump status variables, the nonconvex head difference equations (3), (6), and the pump power consumption equation (4).

C. Incorporating ICNNs

The nonconvex optimal water pumping problem is computationally intensive, making real-time applications difficult. This is mainly due to the nonconvex pipe head loss (6) and pump head gain (3) equalities which have quadratic flow rate terms. To address this, we propose using ICNNs to approximate the nonconvex constraints. ICNNs ensure that the output of the neural network is a convex function of the input by incorporating convexity constraints within the neural network training process [8]. In [9], the authors propose using ICNNs to approximate the Weymouth equation within

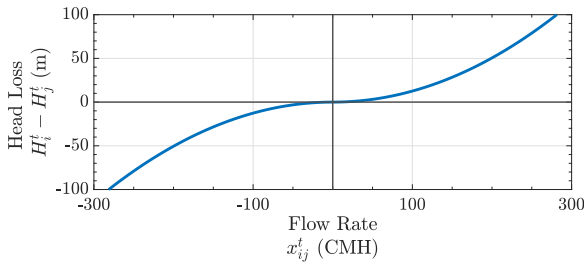


Fig. 1. Pipe head loss function.

the natural gas network. In this work, we investigate how ICNNs extend to the drinking water distribution network, where the Darcy-Weisbach head loss formula shares similarities to the Weymouth equation. However, there are several key differences with the gas flow problem. The pump head gain equation introduces additional nonconvexity through its dependence on quadratic flow rate and pump speed terms as well as the binary pump status variables. To handle these complexities, we combine the ICNN approximation with variable redefinitions and the big-M method. This combination allows us to approximate the nonconvex terms in (6) and (3) as a set of convex constraints. Furthermore, we consider a multiperiod problem. This is necessary because of the inherent time coupling characteristics of water storage tanks.

Fig. 1 illustrates the pipe head loss function (6), where the right side of the equation is convex for nonnegative flow rates and concave for nonpositive flow rates. The term $x_{ij}^t \cdot |x_{ij}^t|$ makes the pipe head loss equality nonconvex and so we replace it with auxiliary variables

$$y_{ij,+}^t + y_{ij,-}^t = \frac{1}{k_{ij}} (H_i^t - H_j^t) \quad \forall ij \in \mathcal{W}, t \in \mathcal{T}, \quad (12)$$

where auxiliary variables $y_{ij,+}^t$ and $y_{ij,-}^t$ represent the convex and concave component.

For the pump equations, we first eliminate the quadratic pump speed terms in both the head gain (3) and pump power (4) constraints. We define a new variable $W_{ij}^t := (w_{ij}^t)^2$ to represent the pump speed squared. This allows us to rewrite (1), (3), and (4) as

$$\underline{w}_{ij}^2 \cdot s_{ij}^t \leq W_{ij}^t \leq \overline{w}_{ij}^2 \cdot s_{ij}^t \quad \forall ij \in \mathcal{P}, t \in \mathcal{T}, \quad (13)$$

$$s_{ij}^t = 1 \Rightarrow H_j^t - H_i^t = b_{ij}^0 W_{ij}^t + b_{ij}^1 x_{ij}^2 \quad \forall ij \in \mathcal{P}, t \in \mathcal{T}, \quad (14)$$

$$p_{ij}^t = c_{ij}^0 \cdot W_{ij}^t + c_{ij}^1 \cdot x_{ij}^t \quad \forall ij \in \mathcal{P}, t \in \mathcal{T}. \quad (15)$$

However, the pump head gain function (14) still contains a quadratic flow rate term on the right side of the equation. We address this using a similar approach to the pipe head, introducing auxiliary variable $y_{ij,+}^t$

$$s_{ij}^t = 1 \Rightarrow y_{ij,+}^t = \frac{1}{b_{ij}^1} (b_{ij}^0 W_{ij}^t - H_j^t + H_i^t) \quad \forall ij \in \mathcal{P}, t \in \mathcal{T}, \quad (16)$$

where the indicator constraint can be implemented in an optimization solver using the big-M method. Note that we only introduce one auxiliary variable here since the pump flow rate must be nonnegative, i.e., (2).

The auxiliary variables $y_{ij,+}^t$ and $y_{ij,-}^t$ represent convex and concave piecewise linear approximations of the original

nonlinear terms. We obtain these approximations through neural network training, where additional requirements are placed on the weights to ensure that the output remains input-convex or input-concave. We refer the reader to [8], [9] for further details and proofs. This process generates a set of unique supporting hyperplanes, which are characterized by their slopes ($\omega_{ij,+}^t$ and $\omega_{ij,-}^t$) and intercepts ($v_{ij,+}^t$ and $v_{ij,-}^t$). The values of these auxiliary variables can then be determined through linear programs

$$y_{ij,+}^t \in \min_{y_{ij,+}^t} y_{ij,+}^t \quad \text{subject to } \omega_{ij,+}^k \cdot x_{ij}^t + v_{ij,+}^k \leq y_{ij,+}^t \quad \forall k \in \mathcal{H}_{ij}^+, \quad (17)$$

$$y_{ij,-}^t \in \max_{y_{ij,-}^t} y_{ij,-}^t \quad \text{subject to } \omega_{ij,-}^k \cdot x_{ij}^t + v_{ij,-}^k \geq y_{ij,-}^t \quad \forall k \in \mathcal{H}_{ij}^-, \quad (18)$$

where $\mathcal{H}_{ij}^+$ and $\mathcal{H}_{ij}^-$ are the convex and concave set of supporting hyperplanes.

This results in a bi-level optimization problem where the lower-level problems are (17) and (18). Similar to [9], we reformulate the bi-level problem as a single-level optimization problem using Karush-Kuhn-Tucker (KKT) conditions. We introduce Lagrange multipliers $\mu_{ij,+}^{k,t}$ and $\mu_{ij,-}^{k,t}$ as auxiliary decision variables for edge ij , time t , and inequality constraint k within the convex and concave problems, respectively. The convex lower-level problem (17) is reformulated for all edges $ij \in \mathcal{E}$ and periods $t \in \mathcal{T}$

$$\omega_{ij,+}^k \cdot x_{ij}^t + v_{ij,+}^k \leq y_{ij,+}^t \quad \forall k \in \mathcal{H}_{ij}^+, \quad (19)$$

$$(\mu_{ij,+}^{k,t} - 1) \cdot M \leq \omega_{ij,+}^k \cdot x_{ij}^t + v_{ij,+}^k - y_{ij,+}^t \quad \forall k \in \mathcal{H}_{ij}^+, \quad (20)$$

$$\sum_{k \in \mathcal{H}_{ij}^+} \mu_{ij,+}^{k,t} = 1, \quad \mu_{ij,+}^{k,t} \in \{0, 1\} \quad \forall k \in \mathcal{H}_{ij}^+. \quad (21)$$

Similarly, the concave lower-level problem (18) can be reformulated for all pipes $ij \in \mathcal{W}$ and periods $t \in \mathcal{T}$

$$\omega_{ij,-}^k \cdot x_{ij}^t + v_{ij,-}^k \geq y_{ij,-}^t \quad \forall k \in \mathcal{H}_{ij}^-, \quad (22)$$

$$\omega_{ij,-}^k \cdot x_{ij}^t + v_{ij,-}^k - y_{ij,-}^t \leq (1 - \mu_{ij,-}^{k,t}) \cdot M \quad \forall k \in \mathcal{H}_{ij}^-, \quad (23)$$

$$\sum_{k \in \mathcal{H}_{ij}^-} \mu_{ij,-}^{k,t} = 1, \quad \mu_{ij,-}^{k,t} \in \{0, 1\} \quad \forall k \in \mathcal{H}_{ij}^-. \quad (24)$$

D. ICNN-Aided Optimization

The resulting ICNN-aided optimal water pumping problem is a mixed-integer linear program, which can be easily solved with off-the-shelf commercial solvers

$$\begin{aligned} & \text{minimize}_{\mathbf{X}} \quad \sum_{t \in \mathcal{T}} \sum_{ij \in \mathcal{P}} \pi_{ij}^t \cdot p_{ij}^t \cdot \Delta T & (\text{ICNN}) \\ & \text{subject to} \quad (2), (5), (7) - (13), (15) - (16), (19) - (24), \end{aligned}$$

where $\mathbf{X}$ contains the decision variables from (NC), pump speed squared W_{ij}^t (instead of pump speed w_{ij}^t), and the auxiliary variables $y_{ij,-}^t$, $y_{ij,+}^t$, $\mu_{ij,-}^{k,t}$, and $\mu_{ij,+}^{k,t}$. The pump speed can be calculated *a posteriori* from the optimal squared pump speed.

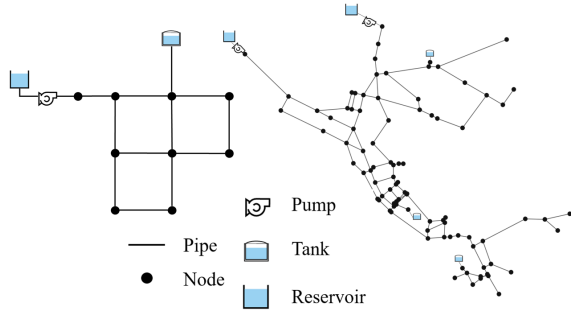


Fig. 2. Water distribution network topology of Net1 (left) and Net3 (right).

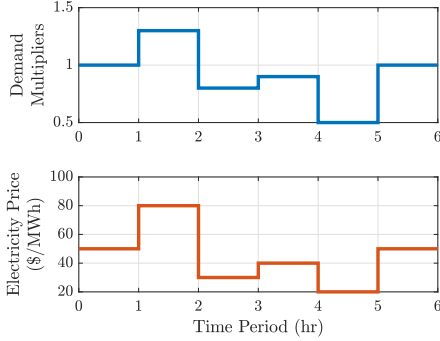


Fig. 3. Water demand multipliers (top) and electricity price (bottom) for Net1.

III. CASE STUDIES

In this section, we investigate the performance of our proposed approach on two case studies. The case study setup is described in Section III-A and our results are presented in Section III-B.

A. Set Up

We evaluate our approach on two example water distribution networks, Net1 and Net3. Fig. 2 illustrates the topology of the networks. These example networks are from EPANET [10], a hydraulic simulator software package. Net1 comprises of 9 junctions, 13 pipes, a reservoir, and a pump. Net3 comprises of 92 junctions, 119 pipes, 2 reservoirs, and 2 pumps. We investigate both networks in order to assess the scalability and computational performance of our approach as we consider larger and more realistic networks.

We solve the optimal water pumping problem for Net1 over six one-hour periods and both the electricity prices and water demands change over time (see Fig. 3).

The optimization problems are solved using the JuMP [11] package in Julia on a 64-bit Intel i7 dual core CPU system running at 2.10 GHz with 32 GB of RAM, where the nonconvex and ICNN-based optimization problems use the IPOPT [12]/Juniper [13] and Gurobi solvers [14], respectively.

B. Results

1) *Comparison of (ICNN) and (NC) Performance:* We first evaluate the cost of our proposed ICNN-based approximation (ICNN) against the original nonconvex water pumping problem (NC) for Net1 over a six-hour scheduling horizon. Since (NC) is a mixed-integer nonconvex problem, the solver

TABLE I
COST COMPARISON (\$)

Problem	Globally optimal	100 Random Min.	Warm Mean	Starts Max.	ICNN Warm Start
(ICNN)	8.759	—	—	—	—
(NC)	—	8.841	8.855	9.035	8.841

TABLE II
SOLVER TIME COMPARISON (SECONDS)

Problem	Globally optimal	100 Random Min.	Warm Mean	Starts Max.	ICNN Warm Start
(ICNN)	1.97	—	—	—	—
(NC)	—	207.81	1,299.04	12,181.28	606.13

may only find a locally optimal solution or may fail to converge altogether. To account for this, we assess the (NC) solution over 100 randomly generated warm start values as well as with the (ICNN) solution as a fixed warm start value. Table I compares the objective values of the (ICNN) and (NC) approach. We found that using the (ICNN) solution as a warm start returned the best (NC) result.

Next, we compare the computation times needed to solve the (NC) and (ICNN) problems for Net1 in Table II. As expected, the solver time for the (ICNN) problem is significantly faster than the (NC) problem, where (NC) takes two to three orders of magnitude longer. This demonstrates that the (ICNN) formulation is better suited for real-time applications. Additionally, the solver time for the (NC) problem varies widely depending on the warm start value. Solving the nonconvex problem took 200 to over 12,000 seconds, depending on the warm start initialization, whereas the (ICNN) problem was solved in around 2 seconds. We also observe that the solver time for the (NC) problem using the (ICNN) solution as a warm start is less than half of the average (NC) solver time over 100 random initializations. This indicates that the (ICNN) solution may be used to find the (NC) solutions faster, where we observed in Table I that the (ICNN) warm start found the best (NC) solution.

C. Feasibility of the (ICNN) Approach

Next, we evaluate the feasibility of the (ICNN) approximation within the original (NC) problem. To do this, we plug the optimal pump control decisions (i.e., pump speed and status) from the (ICNN) solution into the original, nonconvex (NC) formulation and check whether the constraints are satisfied. Within the Net1 case study over a six-hour scheduling horizon, we found that the hydraulic head, water demand, and pump performance constraints were all satisfied. However, we observed that the water level in the tank was slightly lower than we expected, where the final tank level was 0.03% less than than initial tank level. The minor difference in tank level would be negligible, supporting that the (ICNN) formulation is reasonable for solving the water pumping problem.

Fig. 4 illustrates the time-varying pump power consumption across the (ICNN) and (NC) approaches for Net1 over a six-hour scheduling horizon. The (NC) pump power consumption varies depending on the warm start value, as the (NC) problem reaches different local optima based on the initialization. We also observe that the pump power consumption in the (ICNN) solution closely matches the (NC) solution with the (ICNN)

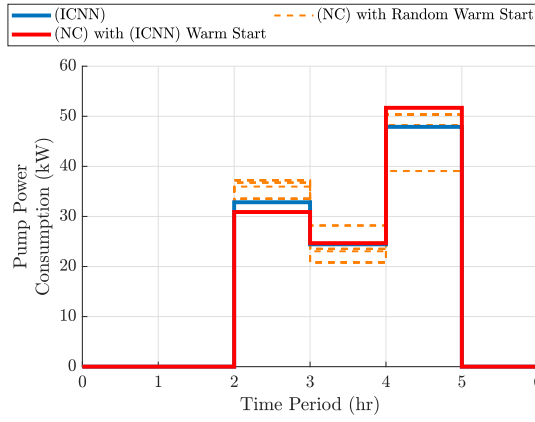


Fig. 4. Pump power consumption over the six-hour scheduling horizon for the (ICNN) solution as well as the (NC) solution with 100 random warm start values and the (ICNN) solution as a fixed warm start.

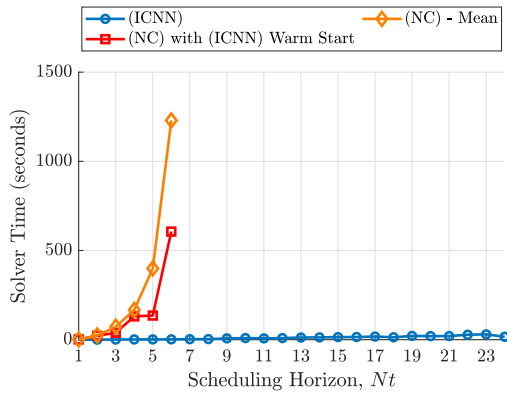


Fig. 5. Solver time as a function of the length of the scheduling horizon. The (ICNN) problem's solver time is compared against the (NC) problem with an (ICNN) warm start and the average solver time of the (NC) problem over 100 random initializations.

warm start, which was the best solution found using the IPOPT/Juniper solver. This suggests that the (ICNN) approach provides near-optimal results within the original, nonconvex water pumping problem while greatly reducing solver time.

D. Impact of Problem Size

Last, we investigate how problem size impacts the computational performance. Increasing the scheduling horizon or network size leads to a bigger optimization problem with more decision variables and constraints. Fig. 5 depicts the solver time for the (ICNN) and (NC) problems over a range of scheduling horizons for Net1. For the (NC) problem, we consider the average solver time over 100 random warm starts. As the scheduling horizon increases, the solver time for the (NC) problem grows exponentially. We limit the (NC) scheduling horizon to six hours since the solver time becomes intractable for longer horizons. In contrast, the (ICNN) approach scales more gradually and offers better computational performance. We also observe that using the (ICNN) solution as a warm start for the (NC) problem generally leads to better than average solver time.

We also test our approach on Net3, which is a larger, more realistic test network. The (NC) approach takes a prohibitively long time to solve, requiring approximately 28.2 hours for a

one-hour scheduling horizon. In contrast, the (ICNN) approach solves in about 15 seconds. Furthermore, using the (ICNN) solution as a warm start for (NC) reduces the solver time to 2.9 hours.

IV. CONCLUSION

In this paper, we formulated the optimal water pumping problem that minimizes the electricity costs given real-time electricity prices and subject to the water flow constraints. The resulting formulation is a nonconvex problem, which is computationally intractable when scaling to realistic networks and scheduling horizons. We propose the use of ICNNs to approximate the nonconvex head difference constraints. ICNNs, along with other reformulation techniques, transforms the problem into a more manageable mixed-integer linear program that can be efficiently solved with commercial solvers, such as Gurobi. In case studies, we found that the ICNN-aided approach offers better performance by efficiently approximating the nonconvex constraints, leading to faster convergence while still finding near-optimal solutions. In the future, we plan to further explore the ICNN-aided approach on larger water networks and against relaxation and approximation techniques proposed in the literature, such as iterative mixed-integer linear programming [3].

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